Industrial Training Logbook Year: 2025–2026

Student's Name: Nada Ashraf Moussa Kamel Gomaa
Student's ID: 120210358

Declaration

Before submitting this document for assessment, please complete the following declaration:

This is to certify that the contents of the Industrial Training Logbook are a true and accurate reflection of the work done by the student in the training company.

Student's name: Nada Ashraf Moussa Kamel Gomaa
Student's signature: Nada Ashraf Moussa Kamel Gomaa
Supervisor's name: Egypt-Digital-Pioneers-Initiative
Supervisor's signature: DEPI- Dr. Eman Raslan
Company's name: CLS- Learning Solutions

Student's Details

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Placement Details

Name of Company: Egypt Digital Pioneers Initiative (CLS)
Company Address: 26 Sharm El-Dokki Main Street, Next to Misr Gas Station in front of Dokki Metro Station, Dokki, Giza Governorate
Industrial Supervisor: CLS learning solutions
Company Contact Tel No.: +20 11 53615270 **Fax No.:** +20 11 53615270
Training Period: 7 Months (April to October 2024)
Allowance: –
(Indicate the amount of allowance paid per month)

Student's Logbook

Training intensity assumption used for planning: 3–4 meetings per week (3–4 hours each) + continuous labs and self-study.

Student’s Logbook	
Date:	1 April – 14 April 2024
	Weeks 1–2
Activities:	• Orientation to the training structure and deliverables (logbook discipline, weekly repo and lab submission workflow). • Orientation to Microsoft SQL and relational database fundamentals. • Learning SQL essentials: DDL (CREATE/ALTER/DROP) and DML (SELECT/INSERT/UPDATE/DELETE). • Hands-on: database setup and environment configuration for practice. • Hands-on: writing simple queries with filters, sorting, grouping, and joins. • Short exercises in query debugging, fixing syntax/logic errors, and validating outputs sample data.
Summary:	Initial phase focused on SQL foundations and establishing a correct working environme followed by intensive practice on essential query patterns required for data engineering analytics workflows.
Key Topics:	• Relational databases and SQL Server environment setup • DDL vs DML and correct usage • Core SELECT queries, WHERE, ORDER BY, GROUP BY • JOIN types (INNER/LEFT/RIGHT) and practical use cases • Query troubleshooting and result validation
Achievements of the Day:	Functional Skills: Computer Skills, Problem-solving Skills Soft Skills: Co

Functional Skills	**Soft Skills**
• Computer Skills • Problem-solving Skills	• Communication Skills • Teamwork Skills

Student's Logbook	
Date:	15 April – 28 April 2024
	Weeks 3–4
Activities:	• Advanced SQL practice: multi-table joins, subqueries, correlated queries, and commo expressions (CTEs). • Data quality logic in SQL: NULL handling, constraints awareness, and consistency ch • Performance basics: understanding indexes at a conceptual level and why queries slov • Applied tasks: create tables + insert sample data + write analytical queries that sim business reporting. • Documentation habit: writing query comments, assumptions, and expected outputs f reproducibility.
Summary:	Deepened SQL capability from basic querying to more complex analytical and engineering-oriented patterns, emphasizing correctness, validation, and early performanc awareness.
Key Topics:	• CTEs, subqueries, correlated queries • Aggregation with HAVING and multi-step analytics • Data validation logic (NULL, missing values, duplicates conceptually) • Basic query performance considerations and indexing concepts • Reproducible SQL documentation practices

Achievements of the Day: Functional Skills: Organizing/Analyzing Data, Decision-making Ability S

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Decision-making Ability	• Critical Thinking Skills • Teamwork Skills

Student's Logbook	
Date:	29 April – 12 May 2024
	Weeks 5–6
Activities:	• Implementing SQL Data Warehouse concepts: OLTP vs OLAP and why analytics re different schema design. • Dimensional modeling practice: fact tables, dimension tables, and star schema vs sno schema. • Normalization vs denormalization trade-offs and when each is justified. • Hands-on: sketching a small warehouse schema for analytics reporting scenarios. • Designed example KPIs and mapped them to facts/measures and dimensions.
Summary:	Shifted from operational SQL usage to analytical system design: learned how warehouse support reporting performance, consistent metrics, and scalable analytics.
Key Topics:	• Data warehouse fundamentals (OLTP vs OLAP) • Fact vs Dimension tables and grain definition • Star vs snowflake schema • Normalization vs denormalization trade-offs • KPI mapping to schema (measures/dimensions)

Achievements of the Day: Functional Skills: Organizing/Analyzing Data, Decision-making Ability S

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Decision-making Ability	• Teamwork Skills • Critical Thinking Skills

Student's Logbook	
Date:	13 May – 26 May 2024
	Weeks 7–8
Activities:	• ETL (Extract, Transform, Load) process concepts and why ETL is essential for wareh reliability. • Data pipeline steps: extracting from sources, cleaning, transforming, and loading into schema. • Data integrity and quality checks: consistency rules, duplicate handling, missing data strategies. • Practical mini-pipeline: simulate ETL with SQL scripts and structured transformatio • Team-based review sessions: comparing schema designs, discussing transformation ru validation logic.
Summary:	Built a structured understanding of ETL pipelines and quality controls required to mai trustworthy analytics datasets in a warehouse environment.
Key Topics:	• ETL workflow and stages • Data quality rules and validation checks • Transformations aligned to business definitions • Loading strategies and schema alignment • Documentation of assumptions and pipeline steps

Achievement of the Day: Functional Skills: Organizing/Analyzing Data, Problem-solving Skills So

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Problem-solving Skills	• Communication Skills • Teamwork Skills

Student's Logbook	
Date:	27 May – 9 June 2024
	Weeks 9–10
Activities:	• Python programming fundamentals for data workflows: syntax, variables, loops, func • Data structures: lists, tuples, dictionaries, sets; selecting structures based on task nee • Error handling with try/except; writing safer scripts. • File handling and directory operations: reading/writing files, working with paths. • Initial work with NumPy arrays and basic Pandas operations for data manipulation.
Summary:	Established Python fundamentals required for automation, data preprocessing, and futu experimentation. Emphasis was placed on correctness, readable code, and controlled err handling.
Key Topics:	• Python basics (functions, loops, conditions) • Data structures and practical usage • Error handling and debugging workflow • File handling and scripting habits • NumPy and Pandas basics

Achievements of the Day: Functional Skills: Computer Skills, Quantitative & Analytical Skills Soft

Functional Skills	**Soft Skills**
• Computer Skills • Quantitative & Analytical Skills	• Communication Skills • Interpersonal Skills

Student's Logbook	
Date:	10 June – 23 June 2024
	Weeks 11–12
Activities:	• Data preprocessing with Python: handling missing values, duplicates conceptually, en categorical data (concept-level). • Exploratory Data Analysis basics: summary stats, simple plots, and interpretation. • Intro to ML workflow idea: data → split → train → evaluate (concept-level). • Strengthened notebook discipline: clean sections, comments, and reproducible runs. • Group sessions: presenting solutions, peer review of code clarity and correctness.
Summary:	Expanded Python skills toward data preparation and analysis patterns that directly sup and data engineering deliverables.
Key Topics:	• Data preprocessing patterns in Python • Basic EDA workflow and interpretation • Reproducible notebook structure and documentation • Clean coding practices and peer review • Intro ML pipeline framing (conceptual)

Achievements of the Day: Functional Skills: Organizing/Analyzing Data, Problem-solving Skills So

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Problem-solving Skills	• Critical Thinking Skills • Teamwork Skills

Student's Logbook

Date:	24 June – 7 July 2024

Weeks 13–14

Activities:

- Introduction to Agile development and Scrum roles/events/artifacts.
- Participation in sprint planning and daily stand-up simulations.
- Understanding tickets/issue tracking systems: breaking tasks, estimating, defining ac criteria.
- Documentation of sprint tasks and aligning technical deliverables with sprint goals.
- Collaboration practice: team communication rules, meeting notes, and action items.

Summary: Learned Agile/Scrum as an operational framework for managing technical tasks, coordi teams, and ensuring traceable progress across complex training modules and project mi

Key Topics:

- Agile principles and Scrum framework
- Sprint planning, daily standups, task breakdown
- Ticketing workflow and acceptance criteria
- Documentation habits for engineering projects
- Collaboration and coordination routines

Achievements of the Day: Functional Skills: Organizing/Analyzing Data, Problem-solving Skills So

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Problem-solving Skills	• Communication Skills • Teamwork Skills

Student's Logbook	
Date:	8 July – 21 July 2024
	Weeks 15–16
Activities:	• Azure Data Fundamentals: core cloud concepts and why cloud matters for data/ML • Understanding cloud storage, compute, and security concepts; shared responsibility concept-level. • Hands-on with Azure Data Studio: connecting, exploring data, and running queries. • Relational vs non-relational databases: use-cases and tradeoffs. • Security and compliance awareness for data systems (concept-level).
Summary:	Developed baseline cloud literacy focused on data platforms, including tool usage (Azur Studio) and fundamental design/security considerations.
Key Topics:	• Cloud data concepts and service categories • Azure Data Studio workflow • Relational vs non-relational databases • Security/compliance awareness (fundamentals) • Cloud value for business and scalability
Achievements of the Day:	Functional Skills: Computer Skills, Decision-making Ability Soft Skills: C

Functional Skills	**Soft Skills**
• Computer Skills • Decision-making Ability	• Communication Skills • Interpersonal Skills

Student's Logbook	
Date:	22 July – 4 August 2024
	Weeks 17–18
Activities:	• Advanced Azure discussion: designing secure and scalable data solutions (concept-lev • Data ingestion and pipeline thinking: batch vs streaming (concept-level). • Monitoring and optimization mindset for cloud workflows (concept-level). • Team-based case studies: mapping business requirements to data services and pipelin components. • Consolidation quiz sessions and practical review.
Summary:	Moved from fundamental cloud concepts to solution-thinking: how to select services anc pipelines while considering scalability, monitoring, and security.
Key Topics:	• Designing data solutions in Azure (concepts) • Batch vs streaming ingestion (overview) • Monitoring and optimization mindset • Security and scalability considerations • Translating requirements to architecture

Achievements of the Day: Functional Skills: Organizing/Analyzing Data, Decision-making Ability S

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Decision-making Ability	• Leadership Skills • Communication Skills

Student's Logbook	
Date:	5 August – 18 August 2024
	Weeks 19–20
Activities:	• Course 7: NLP scope and pipeline: text acquisition, cleaning, normalization, tokeniza modeling, evaluation. • NLU vs NLG: understanding vs generation tasks and examples. • NLP applications covered: toxicity classification, topic classification, grammatical err correction, information retrieval, question answering. • Key NLP challenges: ambiguity (lexical/syntactic/semantic), context dependence, no data, multilinguality/low-resource languages, implicit meaning (sarcasm/cultural con • Preprocessing toolkit concepts: lowercase tradeoffs, regex concept practice, tokenizati (character/word/sentence/subword). • Tool comparison: NLTK vs spaCy (use-cases, tokenization support, pipeline differenc
Summary:	Built the foundations of NLP workflow and challenges, linking preprocessing decisions t downstream model performance and evaluation reliability.
Key Topics:	• NLP pipeline and task taxonomy (NLU/NLG) • Text cleaning and normalization decisions • Tokenization types and trade-offs • NLTK vs spaCy overview • Real-world NLP application mapping

Achievements of the Day: Functional Skills: Problem-solving Skills, Quantitative & Analytical Skills

Functional Skills	**Soft Skills**
• Problem-solving Skills • Quantitative & Analytical Skills	• Critical Thinking Skills • Teamwork Skills

Student's Logbook	
Date:	19 August – 1 September 2024
	Weeks 21–22
Activities:	• Feature engineering for text: one-hot encoding (limitations), Bag of Words, TF-IDF, N-grams. • Understanding sparsity/dimensionality problems and why vocabulary control matters • Traditional ML modeling for NLP: logistic regression, naive Bayes, decision trees (co workflow). • Practice workflow: vectorizer → model → evaluation (concept-level). • Lab-style discussion based on supervised classification pipelines (e.g., fake news-style classification as a practice theme).
Summary:	Advanced from preprocessing to classical NLP representations and traditional ML mod emphasizing interpretability, pipeline structure, and evaluation discipline.
Key Topics:	• One-hot, BoW, TF-IDF, N-grams • Sparsity and dimensionality concerns • Classical supervised NLP models • Vectorizer-to-classifier pipelines • Evaluation framing and error analysis mindset
Achievements of the Day:	Functional Skills: Organizing/Analyzing Data, Decision-making Ability S

Functional Skills	**Soft Skills**
• Organizing/Analyzing Data • Decision-making Ability	• Teamwork Skills • Critical Thinking Skills

Student's Logbook	
Date:	2 September – 15 September 2024
	Weeks 23–24
Activities:	• Deep learning NLP evolution: embeddings and sequence modeling. • Embeddings: Word2Vec (CBOW/Skip-gram), GloVe, FastText; contextual embeddin concept: ELMo/BERT/GPT/T5. • RNN fundamentals and gradient issues; motivation for LSTM and GRU. • Seq2Seq concept: encoder/decoder, bottleneck problem, and attention mechanism co • Evaluation metrics overview: BLEU (translation), ROUGE (summarization), WER (• Transformers overview and why attention is important for long-range dependencies.
Summary:	Covered deep-learning NLP foundations and the motivation for transformer architecture model families to tasks and evaluation metrics.
Key Topics:	• Static vs contextual embeddings • RNN/LSTM/GRU concept and limitations • Seq2Seq and attention concept • BLEU/ROUGE/WER evaluation overview • Transformer motivation and attention role

Achievements of the Day: Functional Skills: Quantitative & Analytical Skills, Problem-solving Skills

Functional Skills	**Soft Skills**
• Quantitative & Analytical Skills • Problem-solving Skills	• Leadership Skills • Communication Skills

Student's Logbook

Date: 16 September – 29 September 2024

Weeks 25–26

Activities:

- Course 8: Hugging Face overview and ecosystem components: Hub, Transformers, Da Tokenizers, Spaces, Accelerate.
- Hub usage concepts: model hub, dataset hub, model/dataset cards, hosted widgets fo testing.
- Transformers library: pre-trained model families and tasks (classification, generation, translation, NER, summarization).
- Pipelines abstraction: tokenization + inference + post-processing as an end-to-end w
- Model architecture families mapping:
  - Encoder: BERT/DistilBERT/RoBERTa for classification/NER/extractive QA.
  - Decoder: GPT-style for text generation.
  - Encoder-decoder: T5/BART/mBART/Marian for translation/summarization/gene QA.
- Discussed transformer challenges: language dominance (English), data availability, long-document cost, opacity, bias in internet-trained models.

Summary: Established working knowledge of Hugging Face tooling and how its libraries and Hub a experimentation, sharing, and deployment for transformer-based workflows.

Key Topics:

- Hugging Face Hub (models/datasets/spaces) and cards
- Transformers library and pipelines
- Datasets and Tokenizers libraries
- Model family selection by task
- Known transformer limitations: bias/opacity/long context

Achievements of the Day: Functional Skills: Computer Skills, Organizing/Analyzing Data | Soft Skil

Functional Skills	Soft Skills
• Computer Skills • Organizing/Analyzing Data	• Communication Skills • Teamwork Skills

Student's Logbook	
Date:	30 September – 13 October 2024
	Weeks 27–28
Activities:	• MLOps definition: practices to manage ML development, deployment, maintenance w automation across the lifecycle. • Why MLOps: scalability, reproducibility, fast time-to-market, and monitoring. • Core data governance concepts: data lineage, data provenance, metadata. • CI/CD in MLOps: automated testing of code and models, version control for code/data/models, automated data validation. • Continuous Training (CT): retraining when data profile changes and drift affects perf • MLflow overview: Tracking, Projects, Models, Model Registry. • MLflow Tracking: logging parameters, metrics, artifacts and using UI for comparison • MLflow Projects: packaging code + environment for reproducibility (Conda/Docker c • MLflow Models: packaging and deployment options (local, REST, cloud/container co • MLflow Model Registry: versioning + staging + production + archived lifecycle cont
Summary:	Built MLOps literacy centered on MLflow as a lifecycle management tool, connecting g and CI/CD/CT concepts to reliable ML delivery.
Key Topics:	• MLOps lifecycle and automation rationale • Data lineage/provenance/metadata • CI/CD adaptations for ML + Continuous Training • MLflow Tracking/Projects/Models/Registry • Reproducibility and versioning practices
Achievement of the Day:	Functional Skills: Quantitative & Analytical Skills, Problem-solving Skills

Functional Skills	**Soft Skills**
• Quantitative & Analytical Skills • Problem-solving Skills	• Critical Thinking Skills • Teamwork Skills

Student's Logbook	
Date:	14 October – 27 October 2024
	Weeks 29–30
Activities:	• Course 9 (Prompt Engineering): prompt structure, constraints, format control, examp (zero-shot vs few-shot), iterative refinement and evaluation prompts. • Applied prompt engineering for practical work: debugging assistance, refactoring note generating checklists and edge cases, improving documentation wording and structure • Final applied project: **cheXray_Classifier** (Pneumonia vs Normal) using Kaggle dat – Dataset: 5,863 pediatric chest X-rays; folders train/test/val with NORMAL/PNEU – Context and reference paper link included in resources. • Colab pipeline implementation (end-to-end): – Kaggle API setup: upload kaggle.json, permissions, download + unzip dataset. – Verified counts per split and label folder; validated directory structure. – Built file path lists using os.walk; shuffled; confirmed lengths. – Preprocessing and augmentation with ImageDataGenerator: rescale, rotation, shift zoom, horizontal flip, fill mode. – Generators via flow_from_directory with target_size=(224,224), batch_size=32, class_mode=binary. – Visual sanity checks: plotting sample batches from train/val/test generators. • Modeling work (baseline + transfer learning + fine-tuning): – Baseline custom CNN (Functional API): Conv2D blocks (16/32/64), BatchNorm, I MaxPool, Dropout, Flatten, Dense, Sigmoid. – Callbacks: ModelCheckpoint (best weights), EarlyStopping, ReduceLROnPlateau. – Transfer learning: ResNet152V2 (ImageNet, include_top=False), frozen base, custo (GAP, Dense, BatchNorm, Dropout, Sigmoid). – Fine-tuning: unfreeze base model with controlled layer freezing (e.g., freeze all but layers), careful LR settings, monitoring overfitting. • Deployment workflow (Hugging Face + Gradio): – Hugging Face login (CLI/notebook), repository use, versioning with add/commit/p concept. – Model saving (.h5/.keras) and packaging for inference. – Gradio demo: image upload → resize to 224x224 → normalize → predict → label (Pneumonia/Normal) with threshold 0.5. – Validated end-to-end inference usability. • Final presentation preparation: problem statement, dataset context, preprocessing, m evaluation approach, deployment screenshots/steps, and lessons learned.
Summary:	Completed an end-to-end ML project under an intensive schedule: dataset ingestion, preprocessing, baseline CNN, transfer learning + fine-tuning, and deployment using Hu Face/Gradio, supported by MLOps and prompt engineering practices.
Key Topics:	• Chest X-ray pneumonia dataset structure and clinical context • Keras data generators and augmentation for images • Custom CNN vs transfer learning vs fine-tuning • Callbacks for training control and reproducibility • Hugging Face Hub/Spaces deployment and Gradio UI • Prompt engineering patterns for debugging, documentation, evaluation
Achievements of the Day:	Functional Skills, Problem Solving Skills, Quantitative & Analytical Skills
Functional Skills	**Soft Skills**

Helpful References (Links)

- Kaggle dataset (Chest X-Ray Pneumonia): https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia
- Dataset clinical context / paper page (Cell): http://www.cell.com/cell/fulltext/S0092-8674(18)30154-5
- TensorFlow ImageDataGenerator docs: https://www.tensorflow.org/api_docs/python/tf/keras/preprocessing/image/ImageDataGenerator
- TensorFlow Transfer Learning guide: https://www.tensorflow.org/tutorials/images/transfer_learning
- Keras Callbacks (EarlyStopping / ReduceLROnPlateau / ModelCheckpoint): https://keras.io/api/callbacks/
- Hugging Face main site: https://huggingface.co/
- Hugging Face Hub docs: https://huggingface.co/docs/hub/index
- Hugging Face Transformers docs: https://huggingface.co/docs/transformers/index
- Hugging Face Spaces docs: https://huggingface.co/docs/hub/spaces
- Gradio documentation: https://www.gradio.app/docs
- MLflow official documentation: https://mlflow.org/docs/latest/index.html